# Graph Convolution for Short-Term Road Speed Prediction in Dense City Grids

Anonymous Authors
Affiliation withheld for review

TABLE I
MEAN ABSOLUTE ERROR IN KM/H

| Model | 15 min | 30 min | 60 min |
|---|---|---|---|
| Per-segment LSTM | 4.9 | 6.3 | 8.1 |
| Graph + GRU (ours) | 4.2 | 5.4 | 7.0 |

***Abstract*—Navigation services predict road speed a few minutes ahead from loop detectors and probe vehicles. We model the road network as a graph and combine graph convolution with a recurrent layer, so a jam spreading from one junction informs the forecast on its neighbors. On 1,800 road segments the model reduces the 15-minute error by 14% over a per-segment recurrent baseline.**

***Index Terms*—traffic prediction, graph neural networks, intelligent transport**

## I. Introduction

Congestion travels along roads: a blocked junction slows the segments that feed it a few minutes later. Forecasting each segment on its own ignores this, while treating the city as an image ignores that two adjacent pixels may be streets with no connection.

Graph convolution [1] passes information along edges of a graph, which matches how traffic moves. We combine it with a recurrent layer [2] that tracks how each segment changes over time.

## II. Method

Nodes are road segments and edges join segments that share a junction, directed in the driving direction. Each node carries the last hour of five-minute average speeds. Two graph convolution layers mix each node with its neighbors up to two junctions away, and a gated recurrent layer turns the sequence of mixed features into a forecast for the next 15, 30 and 60 minutes. We trained with Adam [3] on six months of data.

## III. Results

Table I reports the mean absolute error in km/h. The graph model helps most during the onset of congestion, when the per-segment baseline only reacts after the speed has already dropped. The full week-long speed maps for each district follow in Figs. 1–6.

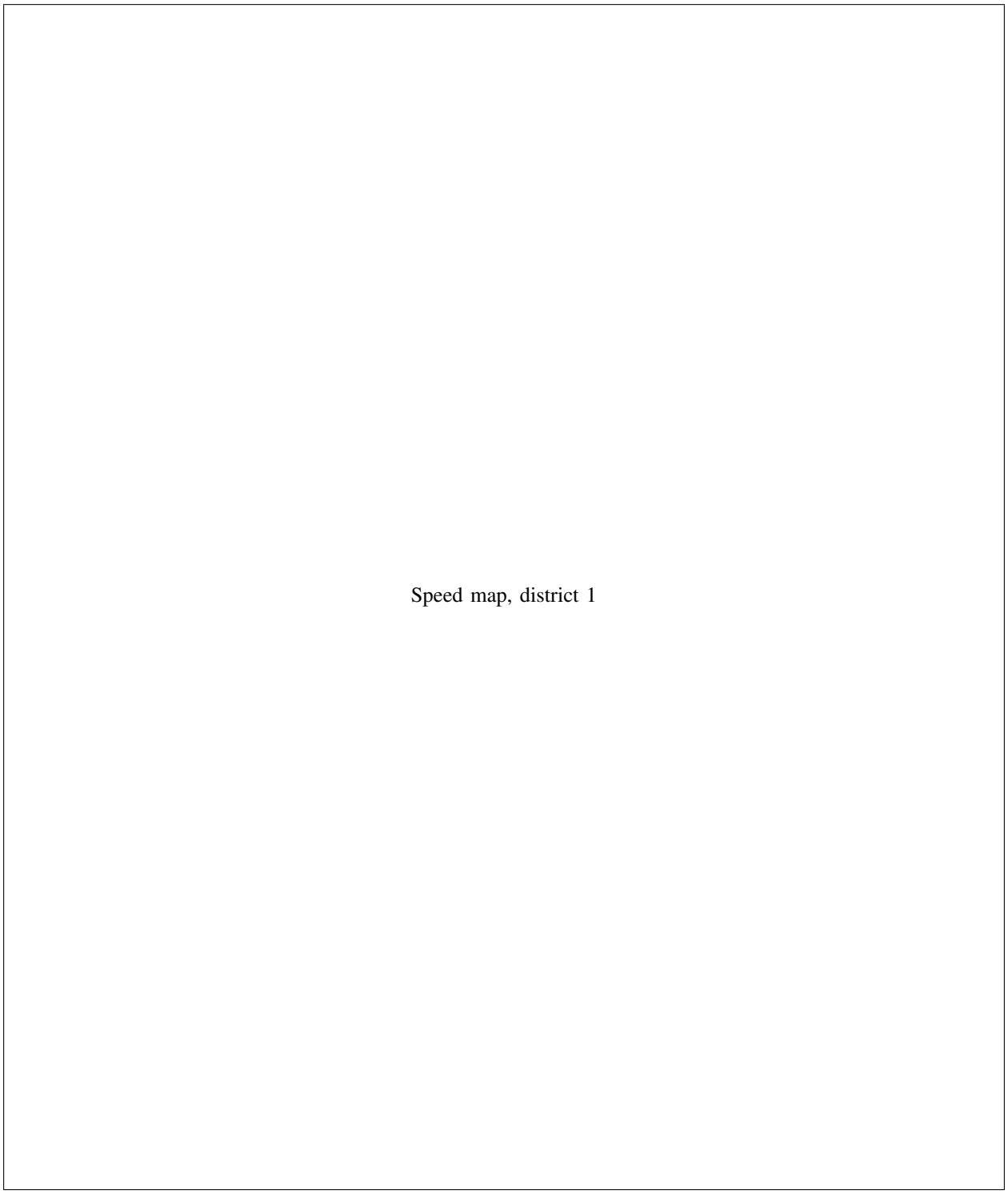



Fig. 1. Predicted and measured speed for district 1 over one week.

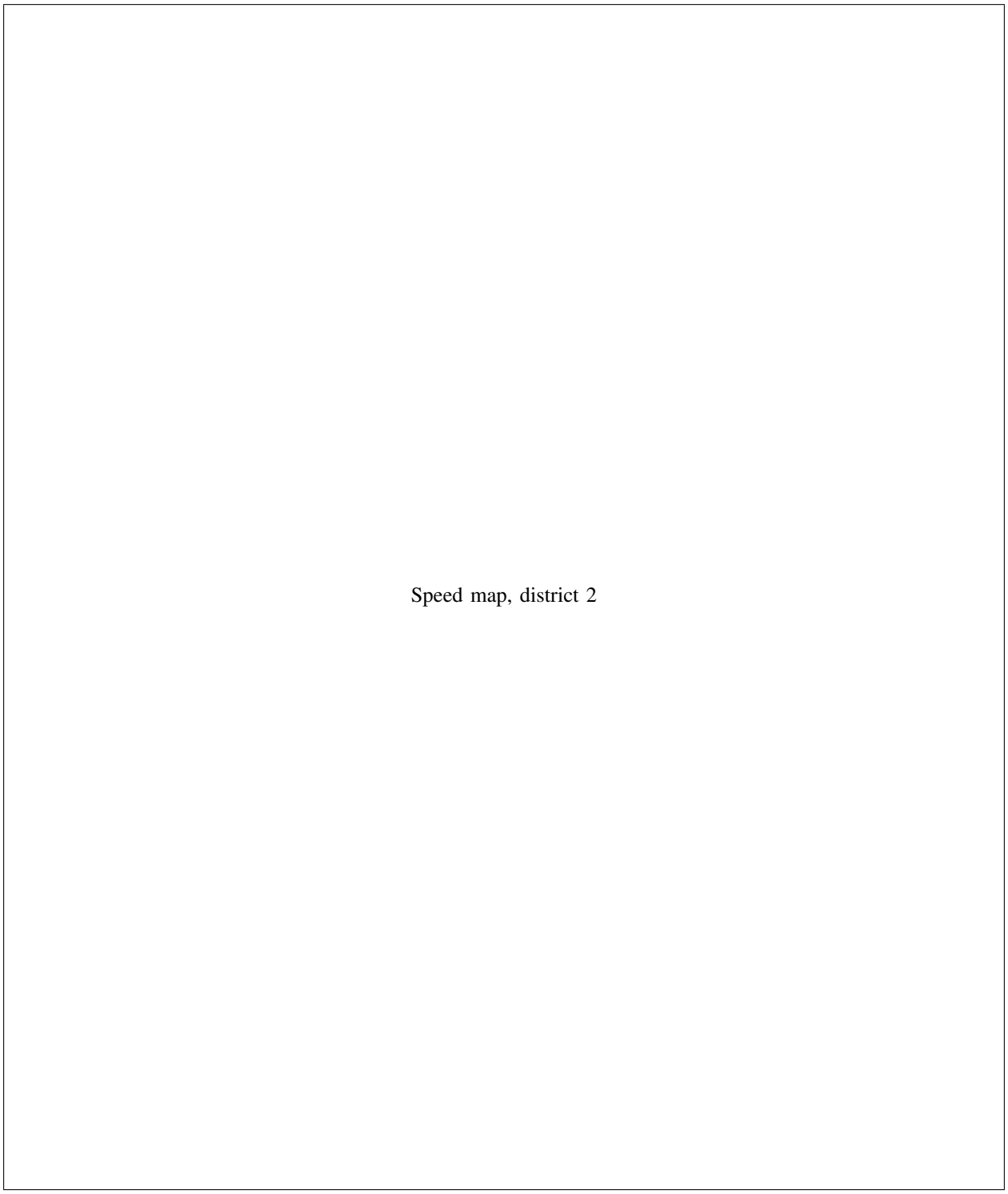



Fig. 2. Predicted and measured speed for district 2 over one week.

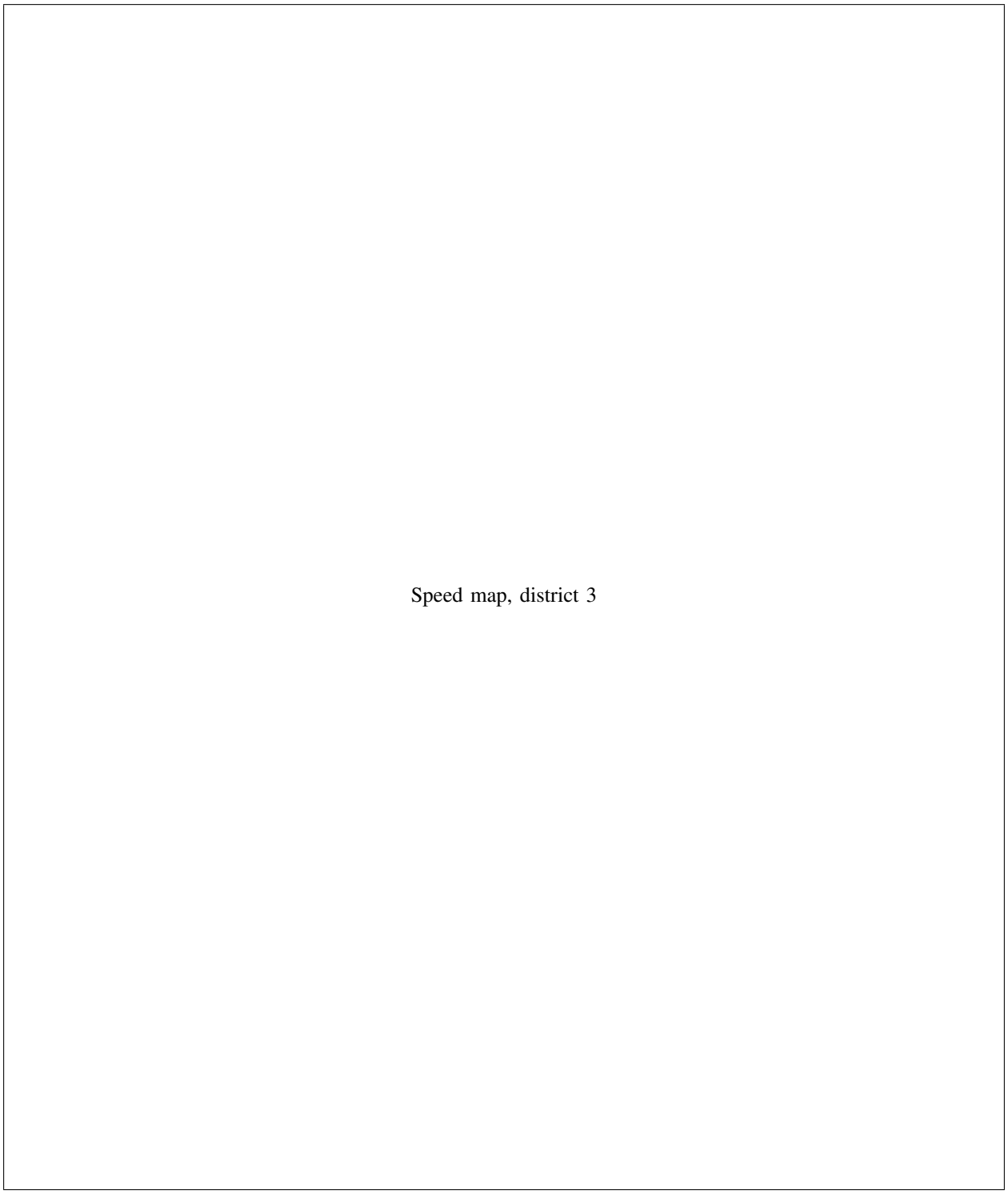



Fig. 3. Predicted and measured speed for district 3 over one week.

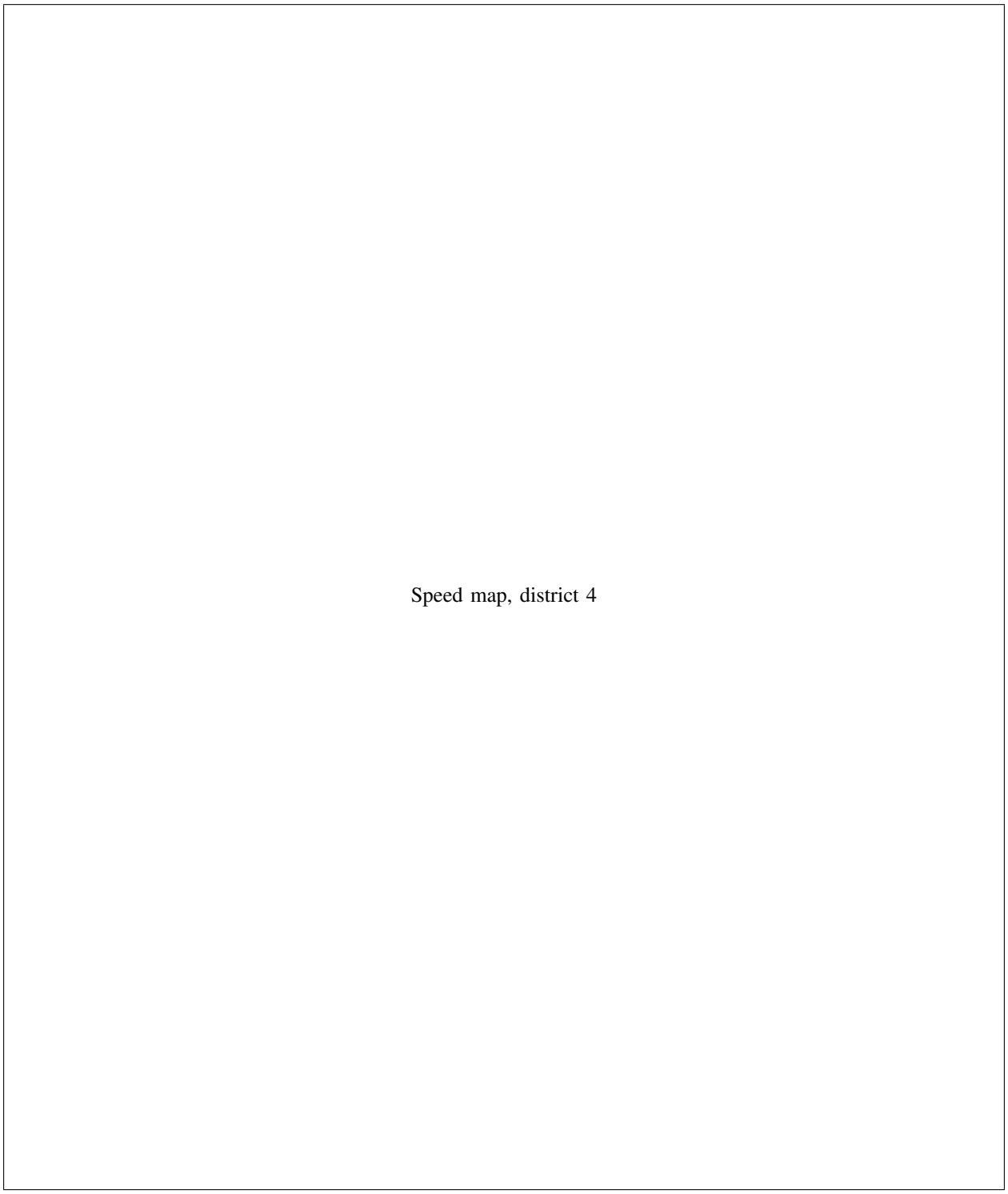



Fig. 4. Predicted and measured speed for district 4 over one week.

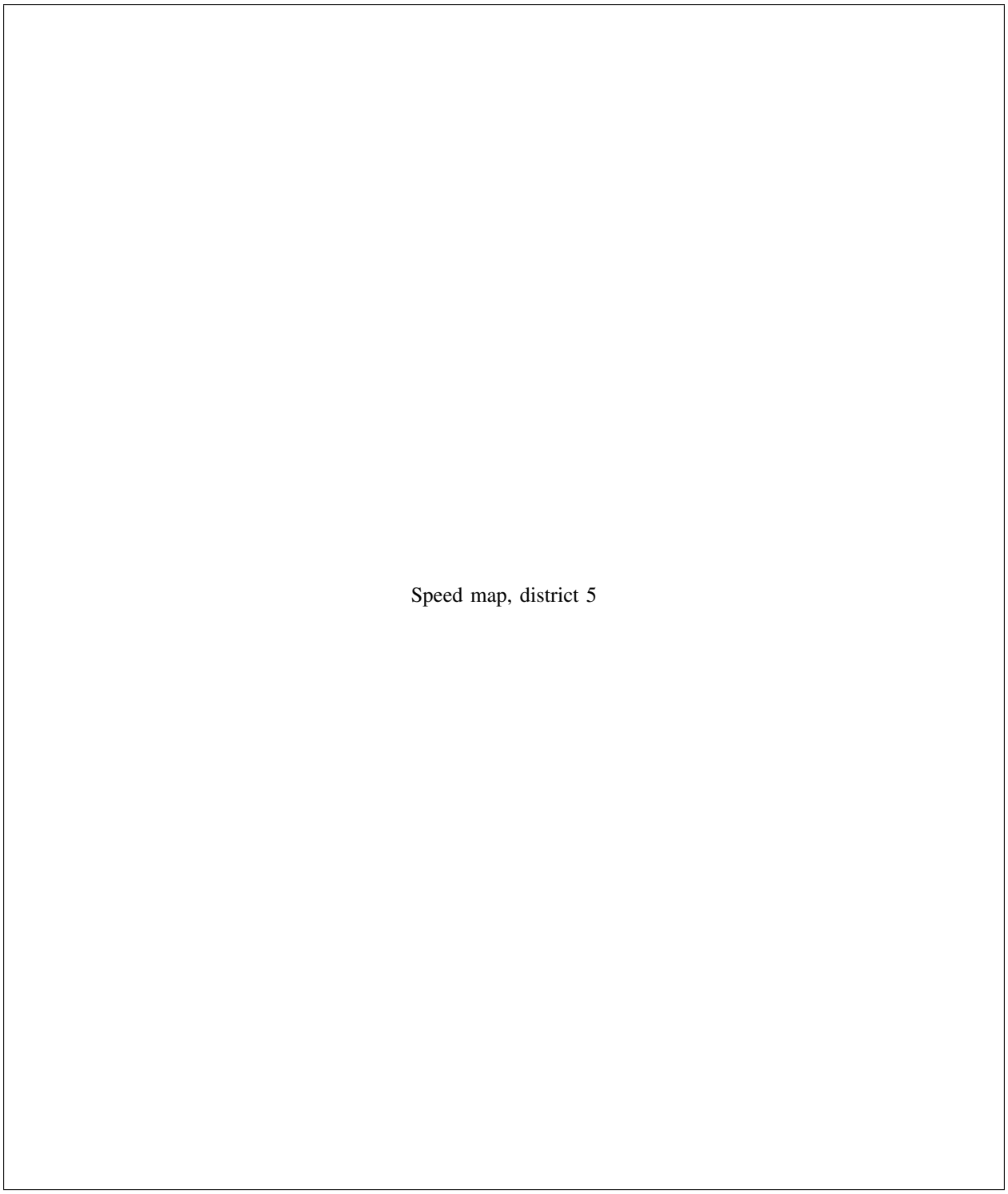



Fig. 5. Predicted and measured speed for district 5 over one week.

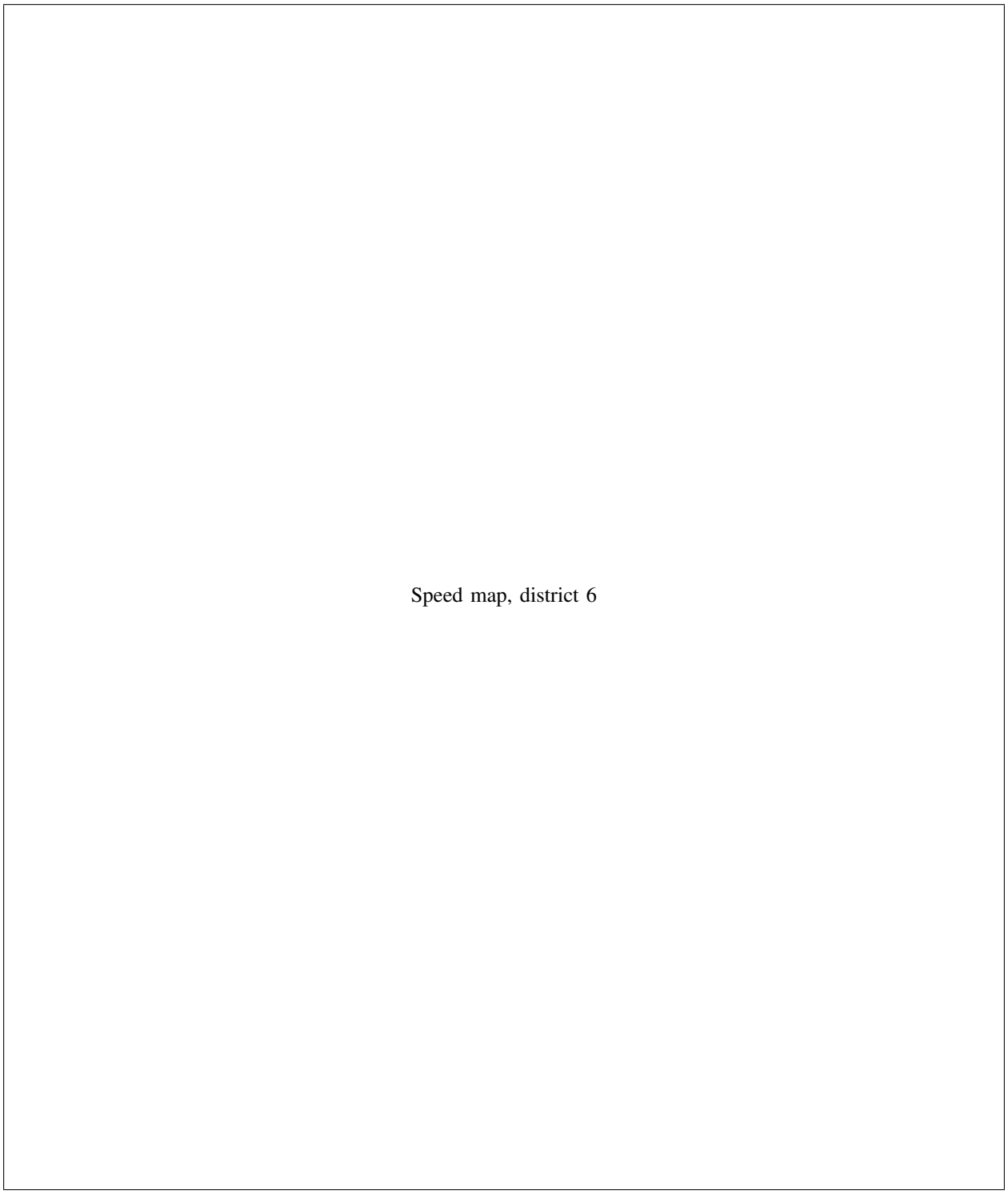



Fig. 6. Predicted and measured speed for district 6 over one week.

## IV. Conclusion

Treating the road network as a graph lets a forecaster see congestion coming from upstream. The gain over per-segment models is largest exactly when drivers need the forecast most, at the start of a jam. Deep learning methods [4] keep improving here, and the graph structure is a cheap, strong prior to build on.

## References


[1] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017.

[2] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[3] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.

[4] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.